SRM TRP Engineering College
Tiruchirappalli
Course Assignment Instructions with Rubrics

Department	Artificial Intelligence And Data Science										
Programme:	B. Tech		**Semester:**	V	**AY:**	2026 - 2027					
Faculty Handling:	Mrs. R. Kalaiyarasi										
Course Code & Title:	AD3501 & Deep Learning										
Assessment Task:			**Assessment Type:**	Term Work/Self ✓		**Required Hrs:**					
Assessment Question based on:	*Open Ended*	✓	*Problem Based Learning*	✓	*Creative Thinking*						
	Close Ended		*Project Based Learning*		*Critical Thinking*	✓					
	Any Other Type	MCQ or any other	*Challenge Based Learning*	✓	*Level of Thinking (RBT)*	K5					
SDG to be Addressed:	SDG is Required	Yes/No	*SDG Number (Specify or Mention Students Choice)*		*Target Number (If Applicable)*						
SDG Integration Level: (If Addressed)	*Level 1: Awareness*		*Level 2: Design Alignment*	✓	*Level 3: Impact and Measurement*						
RIASEC Context	Realistic	Investigative ✓	Artistic	Social	Enterprising	Conventional ✓					
Presentation Format:	PowerPoint/Poster/Video/PDF ✓/Written/Any other (Specify)										
Max Marks:	38										
Relevant Course Outcome(s):											
Submission Deadline:	10/09/26										

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Instructions to the Students:
Assessment Question: scheme guide.
Guidelines: E.g. Guidelines for Preparation/Submission/Expected Output/Review Process/Presentation Format if any

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Rubrics for Evaluation:

Performance Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)

R. Sarulatha - 8147242431 34.

R. Shwetha - 8147242431 59

B. Srisubiksha - 8147242431 51.

38/40

Comprehensive Review of Prior Work: Voice-First, Native-Language Access to Government Welfare Schemes

Prepared in support of: Scheme Guide — A Voice-First, Tamil-Centric Interface for Simplified Access to Government Welfare Schemes

Survey of Prior Work

No.	Paper Title	Author(s)	Journal / Venue	Date	Method / Technology	Strengths	Weaknesses
1	Developing Voice-Based Information Sharing Services to Bridge the Information Divide in Marginalized Communities: A Study of Farmers Using IBM's Spoken Web in Rural India	Chengalur-Smith, Potnis, Mishra	Int. J. Info. Manage.	Apr 2021	Grounded-theory study of an IVR voice platform (IBM Spoken Web) used by Gujarat farmers to record and browse spoken content	Proves illiterate/semi-literate rural users can seek, share, and act on information via ordinary feature phones; identifies a 4-stage model to seed and sustain a voice service	Domain-limited to agricultural knowledge sharing; no NLU/conversational matching; depends on human gatekeepers rather than automated eligibility logic
2	Entertainment Chatbot for the Digital Inclusion of Elderly People Without Abstraction Capabilities	García-Méndez, de Arriba-Pérez, González-Castaño, Regueiro-Janeiro, Gil-Castiñeira	IEEE Access	2021	Hybrid NLG + sentiment-analysis "intelligent radio" chatbot; Spanish speech-to-text/TTS via Google Voice SDK	Explicitly designed for users with low abstraction ability; monitors engagement via sentiment; validated with real elderly and dementia-association participants	Purely entertainment/companionship-oriented, not transactional; no eligibility, document, or application guidance; single-turn dialogue stages only
3	E-Grampanchayat: Empowering Rural Governance with Native Language Technology	Chougule, Shinde, Kamble, Kamble, Devkate	IJARSCT	Feb 2025	NLP + speech recognition + blockchain multilingual e-governance platform (Dialogflow/GPT-based) on a mobile/web app	Voice assistance for non-literate users; pilot reported 60% faster processing and 45% higher access, with elderly and women citing the voice feature as a "game-changer"	Still requires the citizen to know which named service to ask for; dialect variation causes translation errors; digital illiteracy among officials remains a bottleneck
4	Large Language Models Humanize Technology	Kumar, P.	arXiv preprint	May 2023	Conceptual, case-based analysis of LLMs addressing content, tool-complexity, and personalization bottlenecks, illustrated with a Punjabi voice-bot welfare-eligibility example	Articulates why an LLM can answer free-form regional-language questions about entitlements without the user knowing a scheme's name or menu path	Illustrative/conceptual only; no deployed system, empirical user study, or real welfare dataset was tested
5	Understanding Older Adults' Acceptance of Chatbots in Healthcare Delivery: An Extended UTAUT Model	Yu, Chen	Front. Public Health	Nov 2024	Extended UTAUT survey model (PLS-SEM) with 428 Chinese adults aged 60+	Identifies which factors (facilitating conditions, self-actualization, physical condition) drive actual chatbot use, just intention to use,	Survey-based, not an observed interaction study; healthcare-domain only, not welfare/administrative schemes; single-country

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No.	Paper Title	Author(s)	Journal / Venue	Date	Method / Technology	Strengths	Weaknesses
						among elderly users	urban sample
6	Improve Technology and eHealth Literacy in a Retirement Community via the Use of LLM-Powered Chatbot	Chung, Chen, Jeon, Zaslavsky	Innovation in Aging	Dec 2024	Human-centered design + GPT-3.5-based chatbot prototype (Streamlit) with adjustable font, voice input, and simplified navigation	Demonstrates a dual-goal design pattern — information retrieval plus literacy building — with high reported user satisfaction in a pilot	English-only, US retirement-community context; small pilot with no long-term data; text-first design, voice treated only as a future addition
7	Talk2Care: An LLM-Based Voice Assistant for Communication Between Healthcare Providers and Older Adults	Yang, Xu, Yao, Rogers, Zhang, Intille, Shara, Gao, Wang	Proc. ACM IMWUT	Jun 2024	LLM-powered voice assistant paired with a provider-facing summarization dashboard; two interview and usability studies (N=10 older adults, N=9 providers)	Shows an LLM voice interface can reliably collect structured information from older adults conversationally and forward a summary for human review	Built for provider-facing health communication, not welfare-scheme navigation; assumes a human reviewer downstream; English, US context only
8	AI Literacy Education for Older Adults: Motivations, Challenges and Preferences	Tang KangJie, Song, Zhu, Li, Lee	CHI EA '25	Apr 2025	Survey of 103 adults aged 50+ on motivations, learning challenges, and format preferences for AI literacy	Finds older adults want hands-on, low-jargon, one-step-at-a-time learning — directly informing simplified, incremental interaction design	General AI-literacy education focus, not scheme navigation; self-report survey with no interaction with an actual voice system; Singapore-based sample
9	Bridging the Digital Divide Through E-Governance: An Empirical Study of Rural Inclusion and Service Accessibility in Madhya Pradesh, India	Bisen, Bedia	Indian J. Mod. Res. Rev.	Jan 2026	SERVQUAL + TAM structured survey and regression analysis of rural citizens using Common Service Centres (CSCs) across Madhya Pradesh districts	Confirms trust, service quality, and digital confidence — not mere availability — determine e-governance uptake, motivating a low-friction voice-first design	Studies the existing intermediary-based CSC model rather than a personal-device voice system; Hindi/English-medium region, not Tamil-specific
10	Digital Learning Assistant Using Artificial Intelligence for Elderly Health Literacy: A Systematic Review	Wiwatsiripong, Koraneekij, Khlaisang	J. Info. Learn.	2026	PRISMA-2020 systematic review of 17 studies (2011–2025) on AI-based digital learning assistants (chatbots, voice assistants, virtual coaches) for elderly	Synthesizes that culturally/linguistically tailored, co-designed conversational tools following geragogy principles produce the strongest literacy and trust outcomes	Health-literacy domain only, not welfare/administrative schemes; reviewed studies are mostly Western/high-income-country; no India- or Tamil-specific evidence

Identified Gaps in Existing Research

Taken together, the ten papers surveyed above reveal two strands of literature that have matured independently but have never been brought together to address this specific problem.

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- Scheme discovery without free-form language input: the Indian e-governance systems reviewed [1], [3], [9] improve access once a citizen already knows which service or scheme to request, but none accepts an open-ended, natural-language expression of need and maps it to eligible schemes.
- Elderly-conversational-AI design knowledge disconnected from welfare applications: the accessibility and adoption findings from [2], [5], [6], [7], [8], [10] consistently concern health, companionship, or general literacy support — none apply this design knowledge to government welfare-scheme discovery or documentation guidance.
- Absence of Tamil-specific, regional-language evidence: every accessibility-focused study reviewed here is set in Spanish, English (US), Mandarin, or a Singapore English-speaking context; the Indian studies operate in Hindi/English-medium regions, and no reviewed system is built or evaluated for Tamil-speaking elderly users specifically.
- No single pipeline unifying temporary profile construction, audio read-back, and stepwise interaction: each of these capabilities appears in isolation across the ten papers, never combined within one welfare-access system.

These four gaps directly motivate the proposed system's core design: voice- and Tamil-text-based need expression, conversational eligibility questioning, temporary profile construction, simplified explanation, and audio-based reading of essential information, layered on top of — rather than replacing — existing government welfare platforms.

Alignment with Sustainable Development Goals

The reviewed literature and the proposed system align most directly with three United Nations Sustainable Development Goals.

- SDG 1 — No Poverty: Target 1.3 calls for nationally appropriate social protection systems for the poor and vulnerable. Simplifying discovery of, and access to, welfare schemes directly increases the reach and uptake of these protections among elderly and low-income Tamil-speaking citizens.
- SDG 10 — Reduced Inequalities: Target 10.2 calls for empowering and promoting the inclusion of all, irrespective of age. A voice-first, native-language interface reduces the digital and literacy barriers identified across [1], [3], [9], and [10] that otherwise exclude elderly and low-digital-literacy users from public services.
- SDG 16 — Peace, Justice, and Strong Institutions: Target 16.6 calls for effective, accountable, and transparent institutions, and Target 16.10 for public access to information. Reducing dependence on intermediaries and clarifying eligibility, documentation, and official channels supports more transparent and directly accessible public-service delivery.

Bibliography (IEEE Style)

[1] I. Chengalur-Smith, D. Potnis, and G. Mishra, "Developing voice-based information sharing services to bridge the information divide in marginalized communities: A study of farmers using IBM's spoken web in rural India," Int. J. Inf. Manage., vol. 57, art. no. 102283, Apr. 2021.

[2] S. García-Méndez, F. de Arriba-Pérez, F. J. González-Castaño, J. Á. Regueiro-Janeiro, and F. J. Gil-Castiñeira, "Entertainment chatbot for the digital inclusion of elderly people without abstraction capabilities," IEEE Access, vol. 9, pp. 75878–75891, 2021.

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[3] O. A. Chougule, S. S. Shinde, S. P. Kamble, J. S. Kamble, and T. B. Devkate, "E-Grampanchayat: Empowering rural governance with native language technology," Int. J. Adv. Res. Sci. Commun. Technol. (IJARSCT), vol. 5, no. 1, pp. 184–193, Feb. 2025.

[4] P. Kumar, "Large language models humanize technology," arXiv preprint arXiv:2305.05576, May 2023.

[5] S. Yu and T. Chen, "Understanding older adults' acceptance of chatbots in healthcare delivery: An extended UTAUT model," Front. Public Health, vol. 12, art. no. 1435329, Nov. 2024.

[6] R.-Y. Chung, F. Chen, Y. Jeon, and O. Zaslavsky, "Improve technology and eHealth literacy in a retirement community via the use of LLM-powered chatbot," Innov. Aging, vol. 8, suppl. 1, p. 1252, Dec. 2024.

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[8] E. Tang KangJie, T. Song, Z. Zhu, J. Li, and Y.-C. Lee, "AI literacy education for older adults: Motivations, challenges and preferences," in Ext. Abstr. CHI Conf. Human Factors Comput. Syst. (CHI EA '25), Yokohama, Japan, Apr. 2025.

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[10] B. Wiwatsiripong, P. Koraneekij, and J. Khlaisang, "Digital learning assistant using artificial intelligence for elderly health literacy: A systematic review," J. Inf. Learn., vol. 37, no. 1, art. no. e283811, 2026.

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